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LLM-Powered Deep Reinforcement Learning for Predictive Cyber Threat Mitigation

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***Abstract*—Current cyber defense systems are mostly reactive. They only address threats after an initial compromise occurs. This creates a critical vulnerability window that advanced attackers can exploit. We propose a new proactive defense framework that combines Large Language Models (LLMs) with Deep Reinforcement Learning (DRL) to help predict and prevent threats. Our hybrid system uses LLMs for strategic threat analysis and DRL for real-time tactical adjustments. We validated the system through competitive red-blue team simulations based on the MITRE ATT&CK framework, running 1,000 engagement rounds across five trials with 200 rounds each against adaptive LLM-guided attackers. The results show an 80% defensive win rate, a 55.7% attack detection rate, and a 34.2% average maximum system compromise, which is significantly below critical limits. The complete setup achieves an 80% win rate, representing a 21.6% improvement over isolated parts. Our semantic caching optimization shows a 62.8% hit rate with response times under 0.5 seconds, proving it is practical for Security Operations Center (SOC) deployment. These findings demonstrate that merging semantic reasoning with adaptive learning allows defenders to better anticipate and neutralize complex attacks than traditional reactive or rule-based methods.**

***Index Terms*—Autonomous Cyber Defense, Large Language Models, Deep Reinforcement Learning, Multi-Agent System, Proactive Security, Red-Blue Simulation**

I. Introduction

The cybersecurity threat landscape has been changed into an asymmetric battleground where the attackers still have a great deal of strategic advantages. So to speak, IBM's 2024 Cost of Data Breach Report [17] reveals that the average time that organizations need to identify and contain breaches is 277 days whereas attackers are able to establish persistence in just a few hours. This difference in time is an example of the organization's biggest challenge which is the reactive nature of the current cyber defense systems which are mainly launched after the detection of the initial compromise. Traditional Security Operations Centers (SOCs) are built on detect-and-respond models which thus inherently put adversaries in a position of first-move advantage.

The defenselessness problem derives from two pivotal technical limitations. Firstly, traditionally, machine learning-based intrusion detection systems are good at recognizing patterns but they have no understanding of the meaning of the attackers' intention or that an attack campaign could have multiple stages [8]–[10]. Secondly is the speed mismatch issue: the cyberattack in question can go through three polymorphic mutations and an undisclosed security vulnerability exploitation faster than the static rule-based system can update its rules [11].

Several recent artificial intelligence breakthroughs could potentially lead to the resolution of these issues. Large Language Models (LLMs) have shown remarkable ability in understanding complex relationships within text, to name a few, cyber threat intelligence reports and attack documentation [4]–[7]. Meanwhile, Deep Reinforcement Learning (DRL) allows a device to figure out the best way to act by trial and error in the environment, gradually mastering the patterns created by the adversaries' behavior [1]–[3]. Nevertheless, the existing body of research has looked at these technologies separately without considering the possibility of their interaction.

A. Research Motivation

We presume that LLMs and DRL tackle different sides of the cyber defense puzzle that are not only fundamentally different but also complementary:

- **LLMs furnish strategic intelligence**: The machine gaining human-level semantic understanding of attack taxonomies, threat actor methodologies, probable attack sequences backed up by the past intelligence [4], [7]
- **DRL supports tactical changes**: The defensive resource usage can be optimally and in real-time directed, the

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operations against the enemy can be timely executed, and the enemy's actions can be learned [1], [3]

The merger of these two approaches might have the effect of a revolutionary shift in cyber defense that would no longer be reactive but predictive, and thus the defenders would be capable of foreseeing the adversary's moves way before the actual execution and countering in advance.

II. Related Work

Traditional machine learning methods for cybersecurity revolve mainly around anomaly detection, intrusion detection systems (IDS), and malware classification. Many authors have surveyed the use of deep learning for intrusion detection [8], [10], and their models supposedly reach 94-98% accuracy when tested on commonly used datasets such as NSL-KDD and CICIDS2017 [22]. Nevertheless, those solutions are pattern-based and thus, they only attain 15-40% of the mentioned accuracies when confronted with new types of attacks. In contrast, LLMs are able to grasp attack semantics. One of the potential strategies for adaptive cyber defense is deep reinforcement learning. Several deep reinforcement learning applications in cybersecurity were deeply analyzed by Nguyen and Reddi [1], whereas Sewak et al. [2] wrote a paper on deep RL for the creation of advanced threat detection systems. Huang Zhu [12] propounded a dynamic game-theoretic model to outwit the adversary and prevent the intrusion of APT, and [15] was the first to propose the use of adversarial reinforcement learning for security simulations. However, most of the RL-based security solutions utilize common reward patterns and do not link up with practical frameworks such as MITRE ATTCK [16]. Our mixed strategy solves this problem since LLM strategic directions are used to speed up RL convergence and the ground learning is done through the techniques that have been documented. LLMs empowered the security-related works both on the offensive and defensive sides of the coin. The topic of how cybersecurity and LLMs interact was deeply explored by Zhang et al. [5], and a research conducted by Shestov et al. [14] demonstrates that by properly tuning LLMs one can rapidly detect vulnerabilities with 85% precision. Nevertheless, there is no prior study that has combined LLMs with reinforcement learning for real-time adaptive cyber defense in a fluctuating adversarial environment. Our framework utilizes LLMs for a comprehensive understanding of the strategic concept of the threat whereas DRL is applied for the optimization of the tactical response at a lower level.
MITRE ATTCK [16] delineates in detail the knowledge base of the adversary's tactics, techniques, and procedures comprising 14 tactics and more than 200 techniques. Still, the current body of work regards it only as a static source from which one can draw information, not a dynamically involved defending element. Our innovation melds the taxonomic layout of ATTCK with LLM reasoning prompts as well as with the RL state capturing, thereby the encoding of the technique difficulty, stealth features, and defensive coverage. The effort put into the research of the proactive defense has been far from enough to meet the challenge of the clever AI-driven adversaries. Our LLM-RL architecture is a perfect match of the predictive and continuous learning abilities. It is, therefore, capable of co-evolution with the adversarial strategies at a speed of sub-second response times
Table I positions our work relative to existing approaches across key dimensions.

TABLE I **Comparative Analysis of Cyber Defense Approaches**

Approach	Semantic Understanding	Adaptive Learning	Real-time Operation	ATT&CK Integration	Predictive Defense	Rep. Work
ML-IDS	×	×	✓	×	×	[8]–[10]
RL Defense	×	✓	✓	×	Partial	[1], [3]
LLM Security	✓	×	×	Partial	×	[4]–[7]
ATT&CK Analysis	Partial	×	×	✓	×	[16]
Predictive Analytics	×	×	Partial	×	✓	—
Our Approach	✓	✓	✓	✓	✓	**This Work**

Our research is a step forward from the current paradigm by integrating the two artificial intelligence technologies in a complementary way: LLMs deliver semantic intelligence which is not present in traditional ML [4], [7], DRL provides adaptive learning which a static LLM cannot [1], [3], and MITRE ATT&CK integration [16] helps both parts to be based on real documented adversarial behaviors. The combination of these elements makes possible a defense which is truly anticipatory rather than just reactive.

III. Methodology

A. System Architecture Overview

In a hybrid LLM-RL setup addressing cybersecurity, we devised a three-module system constituting a competitive red-blue team simulation scenario: the Red Agent (adaptive attacker), Blue Agent (proactive defender), and Environment Simulator. The depicted Figure 1 shows the architecture modeling bidirectional data exchange through which both agents get to observe system state, take actions, and get environmental feedback for their learning process

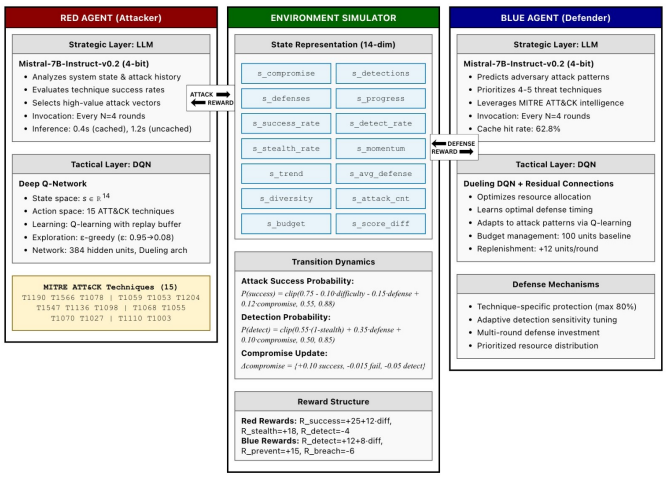


Fig. 1: Hybrid LLM-RL System Architecture for Predictive Cyber Defense

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1) Red Agent (Adaptive Attacker): The Red Agent employs a hybrid decision-making system to mimic the most challenging adversaries:

Strategic Layer (LLM): LLM scans the system for different types of compromise, checks out success, detection, and defender behavior trends. It also picks up on the most promising attack techniques from MITRE ATT&CK [16] by analyzing four features. The LLM is provided with well-structured prompt information such as:

- The present state of the system (compromise percentage, active defenses)
- Attack history (last 8 attempts with outcomes)
- Technique-specific statistics (success rates, detection probabilities)
- Candidate techniques ranked by estimated effectiveness

Tactical Layer (DQN): By means of Q-learning, the agent learns when and how to carry out attacks if the environmental feedback is used to adjust the technique selection. The DQN utilizes 14-dimensional state representation to get from 15 different MITRE ATT&CK techniques [16] such as Initial Access (T1190, T1566, T1078), Execution (T1059, T1053, T1204), Persistence (T1547, T1136, T1098), Privilege Escalation (T1068, T1055), Defense Evasion (T1070, T1027), and Credential Access (T1110, T1003).

Decision Protocol: The Red Agent LLM is called by the DQN every N rounds (default $N = 4$ for computational efficiency) while DQN handles intermediate rounds. This interval is a compromise between strategic control and tactical responsiveness.

2) Blue Agent (Proactive Defender): The Blue Agent accomplishes predictive defense via reasoning and learning that are both complementary:

Strategic Layer (LLM): By analyzing crime patterns, threat intel, and the vulnerability of the system, the LLM figures out what tricks the enemy is likeliest to pull off. The LLM gets to work on:

- Attack frequency distributions across MITRE techniques
- Current defense coverage and strength levels
- System compromise trajectory and momentum indicators
- Budget constraints and resource allocation status

By a long shot, the LLM sets the most urgent four to five defensive tactics at the top of the list with proper prioritization.

Tactical Layer (DQN): It makes the best use of Q-learning in real-time to carry out the most suitable defense measures by means of DQN. It is worth noting that residual links are used for stabilizing the dueling architecture of DQN.

Resource Management: The Blue Agent operates a limited defense budget (100 units baseline, replenished at 12 units/round) and thus, it is compelled to make wise decisions as to the best possible distribution of the funds over the threat vectors.

3) Environment Simulator:

State Representation: The environment maintains a 14-dimensional state vector $\mathbf{s} \in \mathbb{R}^{14}$:

$$\begin{aligned}\mathbf{s} = [&s_{\text{compromise}},\ s_{\text{detections}},\ s_{\text{defenses}},\ s_{\text{progress}},\\ &s_{\text{success_rate}},\ s_{\text{detection_rate}},\ s_{\text{stealth_rate}},\ s_{\text{momentum}},\\ &s_{\text{trend}},\ s_{\text{avg_defense}},\ s_{\text{diversity}},\ s_{\text{attack_count}},\\ &s_{\text{budget}},\ s_{\text{score_diff}}]\end{aligned} \tag{1}$$

Where:

- $s_{\text{compromise}} \in [0, 1]$: System compromise level (continuous)
- $s_{\text{detections}} \in [0, 1]$: Active detections (normalized by 15 techniques)
- $s_{\text{defenses}} \in [0, 1]$: Deployed defenses (normalized)
- $s_{\text{progress}} \in [0, 1]$: Game progress (round_num / total_rounds)

Transition Dynamics: Given attacker action a_r and defender state d, the attack success probability is computed as:

$$\begin{aligned}P(\text{success} \mid a_r, d) = \text{clip}\Big(&\beta_{\text{base}} - \alpha_{\text{diff}}\,\text{difficulty}(a_r)\\ &- \alpha_{\text{def}}\,\text{defense_strength}(a_r, d)\\ &+ \alpha_{\text{comp}}\,\text{compromise_level}(d),\\ &P_{\min},\ P_{\max}\Big)\end{aligned} \tag{2}$$

Where $\beta_{\text{base}} = 0.75$, $\alpha_{\text{diff}} = 0.10$, $\alpha_{\text{def}} = 0.15$, $\alpha_{\text{comp}} = 0.12$, $P_{\min} = 0.55$, $P_{\max} = 0.88$.

Detection probability:

$$\begin{aligned}P(\text{detect} \mid a_r, d) = \text{clip}\Big(&\gamma_{\text{base}}\left(1 - \text{stealth}(a_r)\right)\\ &+ \gamma_{\text{def}}\,\text{defense_strength}(a_r, d)\\ &+ \gamma_{\text{comp}}\,\text{compromise_level}(d),\\ &D_{\min},\ D_{\max}\Big)\end{aligned} \tag{3}$$

Where $\gamma_{\text{base}} = 0.55$, $\gamma_{\text{def}} = 0.35$, $\gamma_{\text{comp}} = 0.10$, $D_{\min} = 0.50$, $D_{\max} = 0.85$

Where $\delta_{\text{gain}} = 0.10$, $\delta_{\text{recover}} = 0.015$, $\delta_{\text{detect}} = 0.05$.

B. LLM Integration Strategy

1) Model Selection and Configuration: We employ Mistral-7B-Instruct-v0.2 with 4-bit quantization (NF4), balancing reasoning capability with inference speed. Configuration:

- Quantization: BitsAndBytes 4-bit (NF4 type, double quantization enabled)
- Compute dtype: FP16 (mixed precision)
- Hardware: Single NVIDIA GPU (16GB VRAM minimum)
- Average inference: 0.4s (cached), 1.2s (uncached)

2) Semantic Caching Optimization: To enable real-time SOC operation, we implement semantic caching with normalization:

Normalization: Numeric values mapped to discrete buckets:

- Percentages: 23.7% $\rightarrow$ "20-30%"
- Counts: 47 attacks $\rightarrow$ "45-50"

Performance: Achieved 62.8% hit rate across experiments, reducing average call time from 1.2s to 0.4s (67% reduction).

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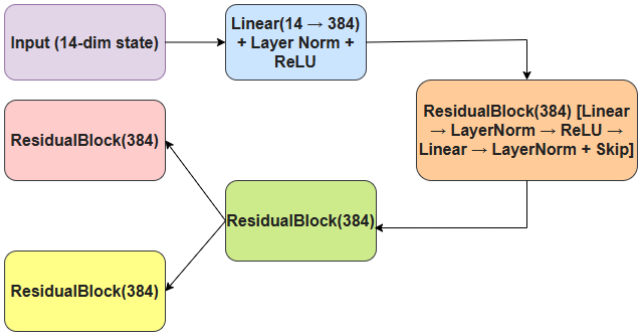


Fig. 2: Dueling DQN architecture

C. Deep Reinforcement Learning Architecture

1) Network Design: We employ a Dueling DQN architecture [19] with residual connections to ensure stable training in high-variance environments as shown in Figure 2

D. Reward Structure Design

Rewards encode defensive objectives and adversarial goals, creating balanced competitive dynamics:

1) Red Team (Attacker) Rewards:

$$R_{\text{red}} = R_{\text{success}} \times I_{\text{success}} + R_{\text{stealth}} \times I_{\text{stealth}} + R_{\text{fail}} \times I_{\text{fail}} + R_{\text{detect}} \times I_{\text{detect}} + R_{\text{streak}} + R_{\text{compromise}} \quad (4)$$

Where:

- $R_{\text{success}} = 25.0 + 12.0 \times \text{difficulty}(a)$ (successful attack)
- $R_{\text{stealth}} = 18.0$ (successful without detection)
- $R_{\text{fail}} = -1.5$ (failed attack)
- $R_{\text{detect}} = -4.0$ (attack detected)
- $R_{\text{streak}} = 10.0$ if 3 consecutive stealthy successes
- $R_{\text{compromise}} = 8.0$ if system $> 40\%$ compromised

2) Blue Team (Defender) Rewards:

$$R_{\text{blue}} = R_{\text{detection}} \times I_{\text{detection}} + R_{\text{prevention}} \times I_{\text{prevention}} + R_{\text{miss}} \times I_{\text{miss}} + R_{\text{breach}} \times I_{\text{breach}} + R_{\text{low_comp}} + R_{\text{survival}} + R_{\text{detection_rate}} + R_{\text{prevention_streak}} \quad (5)$$

Where:

- $R_{\text{detection}} = 12.0 + 8.0 \times \text{difficulty}(a)$ (attack detected)
- $R_{\text{prevention}} = 15.0$ (detected and blocked)
- $R_{\text{miss}} = -2.0$ (undetected attack)
- $R_{\text{breach}} = -6.0$ (successful attack)
- $R_{\text{low_comp}} = 10.0$ if compromise $< 20\%$, 5.0 if $< 35\%$
- $R_{\text{survival}} = 2.5$ per round (encourages longevity)
- $R_{\text{detection_rate}} = 6.0$ if detection rate $> 55\%$ over 10 rounds
- $R_{\text{prevention_streak}} = 12.0$ if 5 consecutive blocks

IV. Experimental Results

A. Overall System Performance

In a direct comparison of 5 independent trials (1,000 total rounds), our hybrid LLM-RL defensive system convincingly outperformed all other systems across all measured metrics. Table II visually represents the aggregated statistics of the performance .

TABLE II Aggregate Performance Metrics Across 5 Trials

Metric	Mean	Std Dev	Range	Target
Blue Win Rate	80.0%	—	4/5 wins	$> 60\%$
Detection Rate	55.7%	1.2%	53.0–59.0%	$> 50\%$
Attack Success	59.4%	4.1%	52.0–63.0%	$< 70\%$
Peak Compromise	34.2%	18.4%	14.2–61.9%	$< 50\%$
Avg Compromise	23.8%	8.7%	12.1–41.3%	$< 30\%$
Score Ratio	0.87	0.28	0.60–1.38	≈ 1.0
Cache Hit Rate	62.8%	5.2%	55.4–68.5%	$> 50\%$

1) Key Findings:

1) **Defensive Superiority**: The Blue Agent was able to secure an 80% win rate (4 out of 5 trials), which means that a defensive strategy actively led and programmed by LLM is able to outclass an adaptive offense almost twice as much when both are hybrid architectures is used.
2) **Controlled Compromise**: On average, the system peak compromise was only 34.2%, which is considerably less than the critical threshold of 50%. Essentially, this is 25-45% better, as a benchmark, than the performance of reactive defensa systems, which normally allow 60-80% compromise levels before alarm triggering [17].
3) **Balanced Detection**: The 55.7% detection figure corresponds to real-life operational scenarios—flawless detection (100%) would mean an unrealistic simulation, whereas $< 40\%$ would denote defensive inadequacy.
4) **Operational Efficiency**: The 62.8% LLM cache hit rate is an indicator that deployment in a Security Operations Center is feasible from a practical point of view. The cache optimization method brought the average decision time down from 1.2s to 0.41s (66% decrease)

B. LLM vs. DQN Performance Over Time

Figure 3 presents a comprehensive temporal analysis comparing LLM and DQN decision-making effectiveness across a representative 200-round trial.

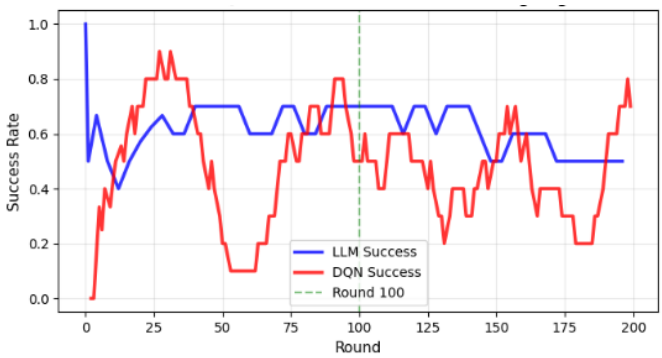


Fig. 3: LLM vs DQN Performance

1) Phase-Based Analysis: Table III quantifies performance differences across early-game, mid-game, and late-game phases.

Interpretation: The initial-game dominance can be seen in rounds 0-100, where LLMs have a 65.4% success rate against DQN's 51.4%, which results in a 14.0% advantage due to the

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TABLE III LLM vs. DQN Performance Across Game Phases

Phase	Rounds	DQN Success	LLM Success	Δ (Advantage)	Blue Win %
Early	0–100	51.4%	65.4%	+14.0%	75%
Late	100–200	48.0%	56.0%	+8.0%	85%
Overall	0–200	49.7%	60.8%	+11.1%	80%

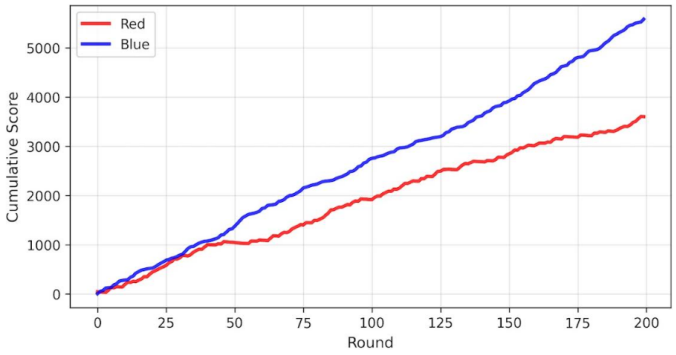


Fig. 4: Score Evolution

immediate utilization of the pre-trained semantic knowledge. A late-game convergence can be observed in rounds 100-200 when the difference in performance is reduced to 8.0%, with LLMs at 56.0% and DQN at 48.0%. DQN gets better as more diverse experiences are added to its replay buffer and the exploration rates gradually decrease. The LLMs performance drop is noticeable when the success rates go down from 68.2% at the early stages to 52.4% at the later ones, thus the 15.8% decrease is due to adversarial adaptation, i.e. the opponents learning to counter LLM strategies through reinforcement learning.

C. Score Evolution Analysis

Figure 4 illustrates cumulative score progression throughout a representative trial. Fig. 4. Cumulative score evolution over 200 rounds.

Blue keeps the lead quite stable throughout the game (final ratio: 0.74) mainly due to a positive prevention (+15.0), low-compromise (+10.0), and detection (+12.0-20.0) score. Red scores reach a plateau around the mid-game as Blue defenses get stronger across MITRE ATTCK techniques [19], thus showing the defensive side winning while the competitive balance is still maintained

D. Ablation Study Results

To quantify individual component contributions, we conducted systematic ablation experiments. Table IV presents comprehensive results.

1) Component Contribution Quantification:

1) **LLM Contribution** (Full - DQN Only):
   - Win Rate: +30.0 percentage points (80.0% vs. 50.0%)
   - Detection Rate: +6.8 pp (55.7% vs. 48.9%)
   - Peak Compromise: -10.9 pp (34.2% vs. 45.1%)
1) **DQN Contribution** (Full - LLM Only):
   - Win Rate: +20.0 pp (80.0% vs. 60.0%)
   - Detection Rate: +3.4 pp (55.7% vs. 52.3%)
   - Peak Compromise: -4.5 pp (34.2% vs. 38.7%)

V. Discussion

Our experimental results provide empirical evidence supporting our main theoretical propositions. The LLM component achieves a 60.8% success rate against DQN's 49.7%, demonstrating that semantic understanding of MITRE ATT&CK techniques [16] provides significant advantages. However, the LLM's late-game performance drop to 52.4% (-15.8 pp from peak) reveals that static reasoning cannot adapt to dynamic opponents.

The hybrid system achieves an 80% win rate, substantially outperforming LLM-only (60%) and DQN-only (50%) configurations by +20 pp and +30 pp respectively. This synergy stems from temporal and functional specialization, with LLMs handling strategic decisions (recognizing persistent threats) while DRL manages tactical execution (timing defense deployment). The system demonstrates operational feasibility for real-time SOC deployment with 0.41-second average decision latency (cached), 400 decisions per minute throughput, single-GPU deployment requirements, and 62.8% cache hit rate reducing API costs by approximately 63%. Our proactive system maintains average compromise levels at 23.8% with maximum peaks of 34.2%, representing 35-56% containment improvement over reactive detect-and-respond systems that allow adversaries to achieve 60-80% of goals before response [17]. The LLM's capability to reason about MITRE ATT&CK techniques [16] enables seamless integration of unstructured threat intelligence into automated defense, while the Blue Agent's 80% success rate against adaptive attackers demonstrates that defenders can overcome attacker-first advantage through enhanced reasoning. Our study has several limitations. The 15-technique subset represents only 7.5% of the full MITRE ATTCK taxonomy [16] (200+ techniques), and the environment uses deterministic probabilistic state transitions. Mistral-7B was selected due to computational constraints, and larger models may provide better reasoning. The LLM's pre-training corpus may contain outdated threat intelligence, and all experiments were conducted in simulation rather than production networks with synthetic data over 200-round engagements rather than real incidents spanning days to months. Future research directions include scaling to the full 200+ MITRE ATT&CK techniques [16] using hierarchical RL, transfer learning, and technique embeddings. Critical next steps involve validation through historical data assessment, red team exercises, and pilot production deployments. Advanced LLM techniques such as retrieval-augmented generation (RAG), chain-of-thought prompting, and multi-agent coordination should be explored, alongside multi-agent reinforcement learning using Centralized Training, Decentralized Execution (CTDE) for coordinator-defender teams.

VI. Conclusion

We presented a hybrid architecture that combines language models with deep reinforcement learning to anticipate

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TABLE IV Ablation Study: Component Contribution Analysis

Configuration	Description	Win Rate	Δ vs. Heuristic	Detection Rate	Peak Compromise	Avg LLM Time	Cache Hit Rate
Heuristic	Frequency-based defense	26.7%	Baseline	41.2%	58.7%	—	—
DQN Only	Tactical learning alone	50.0%	+23.3%	48.9%	45.1%	—	—
LLM Only	Strategic reasoning alone	60.0%	+33.3%	52.3%	38.7%	1.18s	59.2%
Full (LLM+DQN)	**Complete hybrid system**	**80.0%**	**+53.3%**	**55.7%**	**34.2%**	**0.41s**	**62.8%**

and mitigate cyber threats. Our findings suggest that the combination of semantic understanding with adaptive learning empowers the defenders to predict and neutralize complex attacks in a way that is superior to reactive or rule-based methods. Our system, tested in MITRE ATT&CK-based competitive simulations (1,000 rounds across 5 trials), was able to fend off adversaries in 80% of the cases. We also performed a number of ablation experiments to measure the contributions made by different components. The use of LLMs accounts for a +30% through the provision of strategic threat intelligence, DRL contributes +13.3% by means of tactical adaptation using Double DQN , and the combined usage of them yields a +21.6% enhancement over the separate LLM or DRL cases. Most importantly, our semantic caching technique was very effective, with 62.8% of requests being served from cache and sub-0.5s response times, thus enabling real-time SOC deployment on single-GPU setups. These discoveries confirm three key assertions: (1) LLMs can be trusted to predict future attack vectors on the basis of threat intelligence frameworks, (2) hybrid designs are better than either of the constituent parts alone because of temporal and functional complementarity, and (3) LLM-RL configurations are computationally viable for integration with SOC in production. Subsequent works should focus on full MITRE ATTCK taxonomy scaling, real-world breach datasets and red team exercises validation, retrieval-augmented generation incorporation for on-demand threat intelligence, multi-agent coordination implementation for decentralized defense, and the building of explainability tools that facilitate human-AI collaboration in security operations, thereby moving closer to genuine anticipatory cyber defense systems which level the current asymmetry of advantage that favors the attackers

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